

CSE 150A-250A AI: Probabilistic Methods

Lecture 4

Fall 2026

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Slides adapted from previous versions of the course (Prof. Lawrence, Prof. Alvarado, Prof. Berg-Kirkpatrick)

Agenda

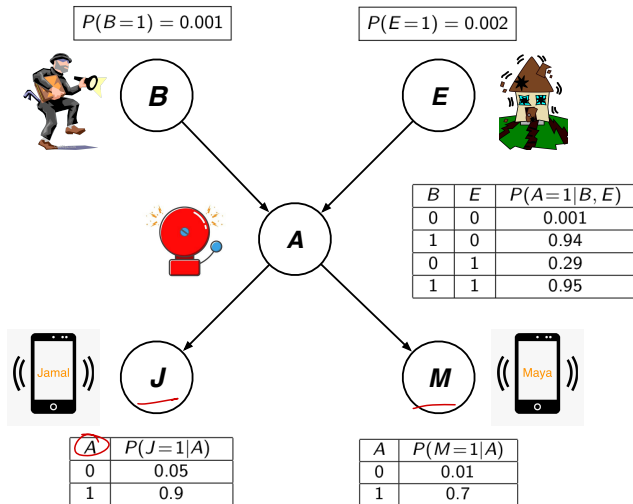
Review

Conditional probability tables

d-separation and examples

Review

Alarm example



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1. Nodes represent random variables.
2. Edges represent (direct) dependencies.
3. Conditional probability tables (CPTs) describe how each node depends on its parents.

$$\text{BN} = \text{DAG} + \text{CPTs}$$

Marginal and conditional independence in DAGs

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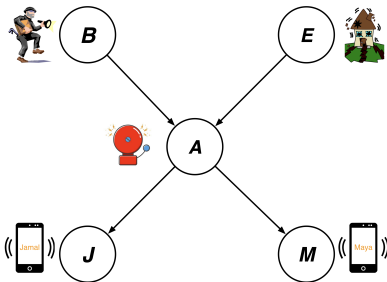
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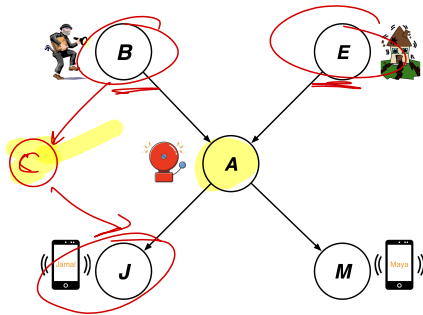
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- Alarm example:



$$P(E) = P(E|B)$$

$$P(J|A) = P(J|A, B, E)$$

$$P(M|A) = P(M|A, B, E, J)$$

Marginal and conditional independence in DAGs

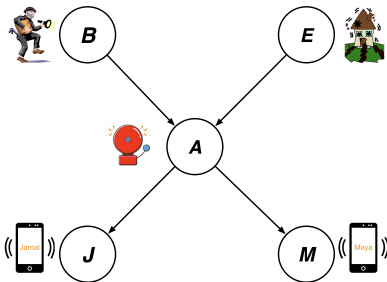
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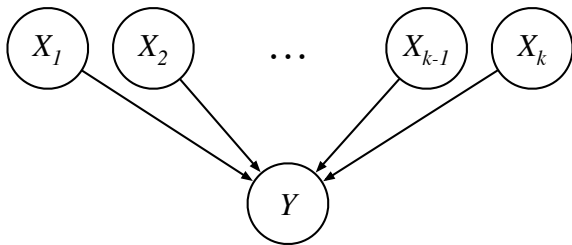
$$P(J|A) = P(J|A, B, E)$$

$$P(M|A) = P(M|A, B, E, J)$$

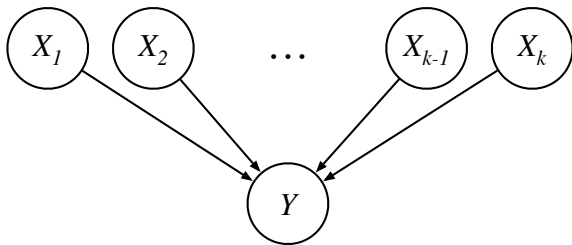
These are true no matter what CPTs are attached to the nodes in the DAG.

Conditional probability tables

Representing CPTs

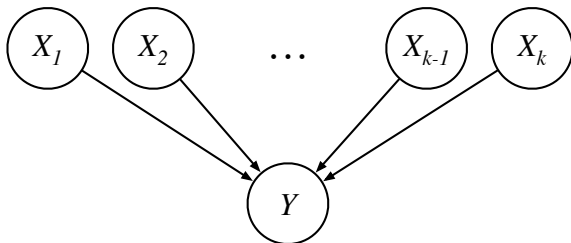


Representing CPTs



- How to represent $P(Y|X_1, X_2, \dots, X_k)$?

Representing CPTs

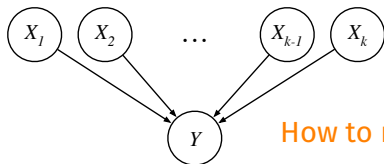


- How to represent $P(Y|X_1, X_2, \dots, X_k)$?
- Simplest case:

Suppose $X_i \in \{0, 1\}$, $Y \in \{0, 1\}$ are **binary** random variables.

How to represent $P(Y=1|X_1, X_2, \dots, X_k)$?

Types of CPTs

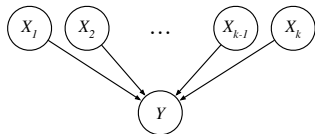


How to represent $P(Y=1|X_1, X_2, \dots, X_k)$?

Some possibilities:

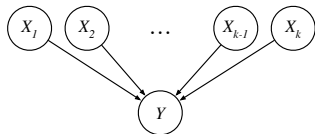
1. Tabular $\longrightarrow$
2. Logical / Deterministic
3. Noisy-OR
4. Sigmoid

1. Tabular CPT



X_1	X_2	$\dots$	X_k	$P(Y=1 X_1, X_2, \dots, X_k)$
0	0	$\dots$	0	0.1
1	0	$\dots$	0	0.6
0	1	$\dots$	0	0.3
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
1	1	$\dots$	1	0.2

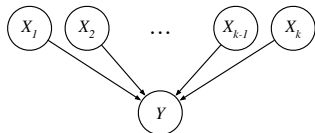
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A lookup table can exhaustively enumerate a conditional probability for every configuration of parents.

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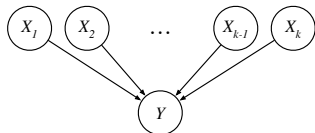
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Pro

Able to model arbitrarily complicated dependence.

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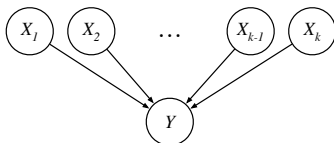
Pro

Able to model arbitrarily complicated dependence.

Con

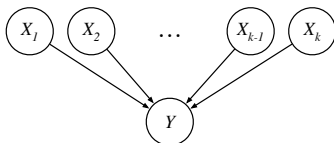
A table with 2^k rows is too unwieldy for large k .

2. Logical / Deterministic CPT



CPTs can also mimic the behavior of logical circuits.

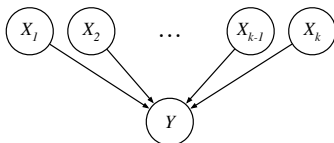
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AND gate

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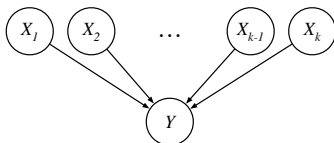


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$$P(Y=1|\underline{X_1}, \underline{X_2}, \dots, \underline{X_k}) = \prod_{i=1}^k X_i$$

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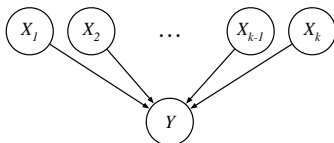
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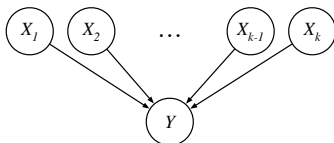
$$P(Y=1|X_1, X_2, \dots, X_k) = \prod_{i=1}^k X_i$$

OR gate

$$P(Y=0|X_1, X_2, \dots, X_k) = \prod_{i=1}^k (1 - X_i)$$

Handwritten red note: $X_i = 1$

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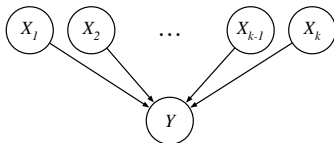
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Pro

Compact representation for large k .

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$$P(\underline{Y=1} | X_1, X_2, \dots, X_k) = \prod_{i=1}^k X_i$$

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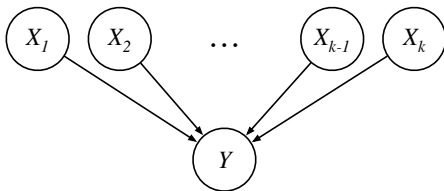
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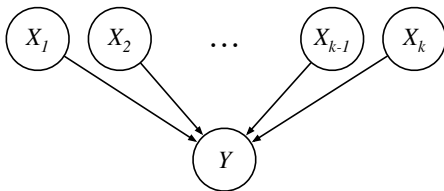
No model of uncertainty.

$$0 < P(Y=1 | x_1, \dots, x_k) < 1$$

3. Noisy-OR CPT

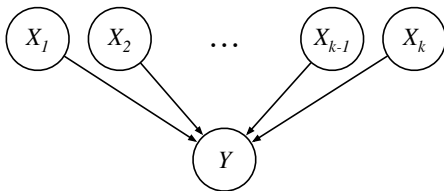


3. Noisy-OR CPT



Use k numbers $p_i \in [0, 1]$ to parameterize all 2^k entries in the CPT:

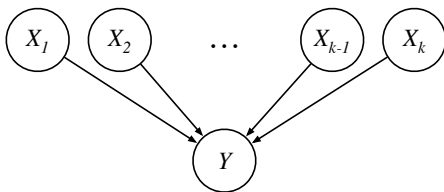
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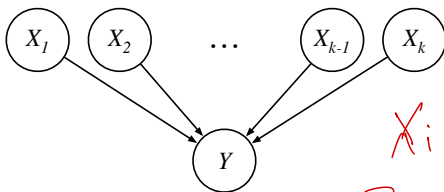


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$$\underline{P(Y=1|X_1, X_2, \dots, X_k)} = \underline{1 - \prod_{i=1}^k (1 - p_i)^{X_i}}$$

3. Noisy-OR CPT



$$X_i \rightarrow 1 / 0$$

$$P_i = 1 / 0$$

Use k numbers $p_i \in [0, 1]$ to parameterize all 2^k entries in the CPT:

$$X_i = 1$$

$$P(Y=0|X_1, X_2, \dots, X_k) = \prod_{i=1}^k (1 - p_i)^{X_i} \quad 0 / 1$$

$$P_i = p_i$$

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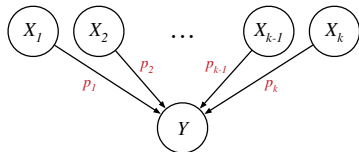
$$X_1 \dots X_{i-1} = 0$$

$$X_{i+1} \dots X_k = 0$$

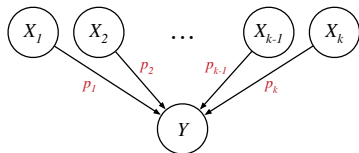
But why is this called Noisy-OR?

$$1 - (1 - p_i) = p_i$$

Noisy-OR CPT (con't)

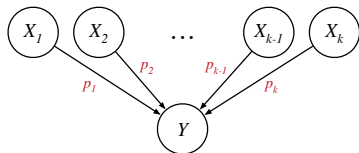


Noisy-OR CPT (con't)



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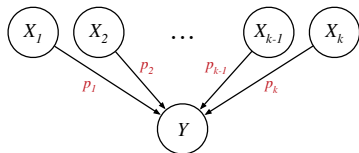
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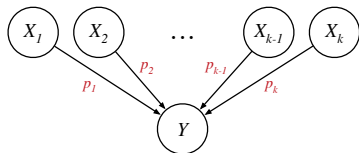


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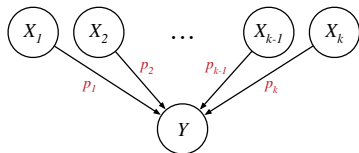


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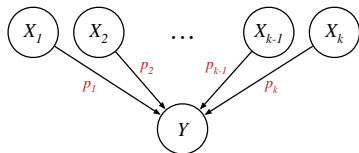


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Noisy-OR CPT (con't)

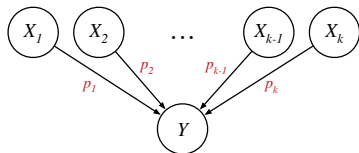


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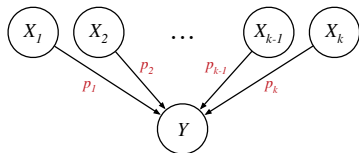
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- When exactly one parent X_j is equal to one:

Noisy-OR CPT (con't)



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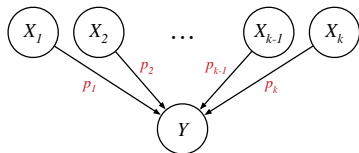
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$$P(Y=1|X_1=0, \dots, X_{j-1}=0, X_j=1, X_{j+1}=0, \dots, X_k=0)$$

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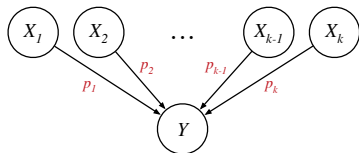
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$$\begin{aligned} P(Y=1|X_1=0, \dots, X_{j-1}=0, X_j=1, X_{j+1}=0, \dots, X_k=0) \\ = 1 - (1 - p_1)^0 \cdots (1 - p_{j-1})^0 \underline{(1 - p_j)^1} (1 - p_{j+1})^0 \cdots (1 - p_k)^0 \end{aligned}$$

Noisy-OR CPT (con't)



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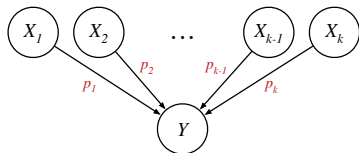
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$$\begin{aligned} P(Y=1|X_1=0, \dots, X_{j-1}=0, X_j=1, X_{j+1}=0, \dots, X_k=0) \\ &= 1 - (1 - p_1)^0 \cdots (1 - p_{j-1})^0 (1 - p_j)^1 (1 - p_{j+1})^0 \cdots (1 - p_k)^0 \\ &= 1 - (1 - p_j) \end{aligned}$$

Noisy-OR CPT (con't)



$$P(Y=1|X_1, X_2, \dots, X_k) = 1 - \prod_{i=1}^k (1 - p_i)^{X_i}$$

- When all parents are equal to zero:

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- When exactly one parent X_j is equal to one:

$$P(Y=1|X_1=0, \dots, X_{j-1}=0, X_j=1, X_{j+1}=0, \dots, X_k=0)$$

$$= 1 - (1 - p_1)^0 \cdots (1 - p_{j-1})^0 (1 - p_j)^1 (1 - p_{j+1})^0 \cdots (1 - p_k)^0$$

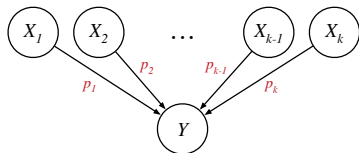
$$= 1 - (1 - p_j)$$

$$= p_j$$

Handwritten red notes and corrections:

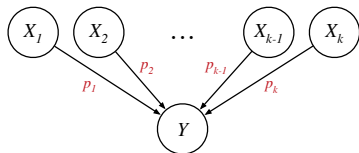
- A red arrow points from the underlined $P(Y=1)$ in the previous equation to the $P(X_i \text{ OR } X_j)$ term.
- A red arrow points from the $1 - (1 - p_j)$ term to the expression $1 - (1 - p_j)(1 - p_i)$.
- A red circle highlights the expression $\bar{p}_i + p_j - p_i p_j$.
- A red checkmark is next to the expression $P(X_i \text{ OR } X_j)$.

Noisy-OR CPT (con't)



$$P(Y=1|X_1, X_2, \dots, X_k) = 1 - \prod_{i=1}^k (1 - p_i)^{X_i}$$

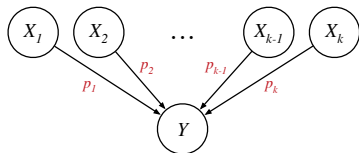
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- Modeling uncertainty

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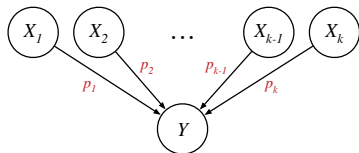


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Intuitively, $p_i \in [0, 1]$ is the probability that $X_i=1$ by itself triggers $Y=1$.

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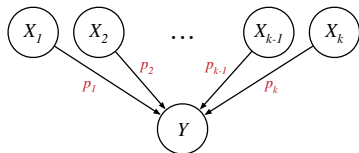
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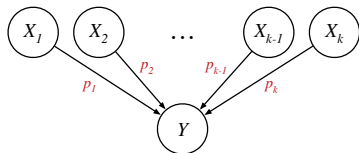
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We recover a logical OR gate by taking the limit $p_i \rightarrow 1$ for all parents $i = 1, 2, \dots, k$.

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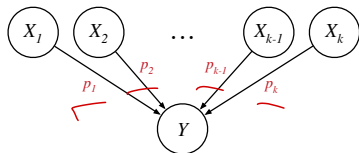
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Noisy-OR CPT (con't)



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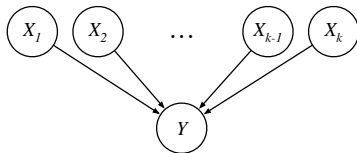
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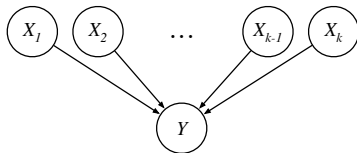
The parents $\{X_i\}_{i=1}^k$ are diseases, and the child Y is a symptom. The more diseases, the more likely is the symptom.

4. Sigmoid CPT



Use k real numbers θ_i $\in \Re$ to parameterize all 2^k entries in the CPT:

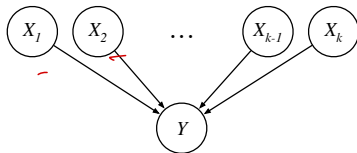
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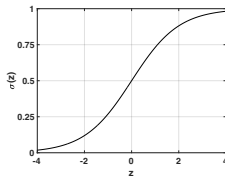


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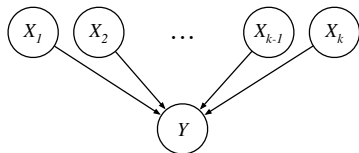
$$P(Y=1|X_1, X_2, \dots, X_k) = \sigma \left(\sum_{i=1}^k \theta_i X_i \right)$$

The function on the right hand side is called the **sigmoid** function:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$



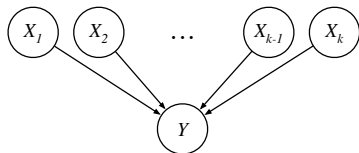
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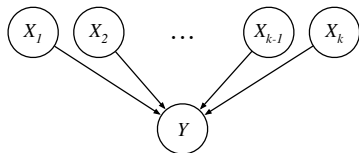


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- Inverse of the link function for logistic regression

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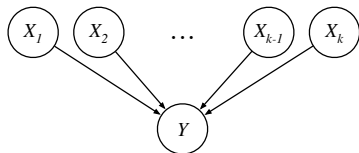
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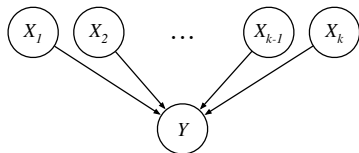
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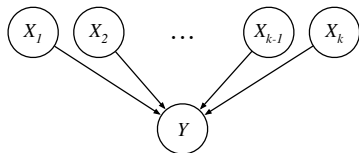
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Properties:

- If $\theta_i > 0$, then $X_i=1$ favors $Y=1$.
- If $\theta_i < 0$, then $X_i=1$ inhibits $Y=1$.
- These effects can mix in a sigmoid CPT (unlike noisy-OR).

d-separation and examples

Conditional independence

- What we've already seen

Conditional independence

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A node X_i is conditionally independent of its non-parent ancestors given its parents:

$$P(X_i | X_1, X_2, \dots, X_{i-1}) = P(X_i | \text{pa}(X_i))$$

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Let X , Y , and E refer to disjoint sets of nodes in a BN.
When is X conditionally independent of Y given E ?

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Let X , Y , and E refer to disjoint *sets* of nodes in a BN.
When is X conditionally independent of Y given E ?

$$\text{When is } \left\{ \begin{array}{lcl} P(X|E, Y) & = & P(X|E) \\ P(Y|E, X) & = & P(Y|E) \\ P(X, Y|E) & = & P(X|E) P(Y|E) \end{array} \right\} ?$$

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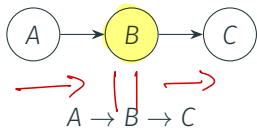
- Above is special case

$$X = \{X_i\}, \quad E = \text{pa}(X_i), \quad Y = \{X_1, X_2, \dots, X_{i-1}\} - \text{pa}(X_i)$$

Base Cases

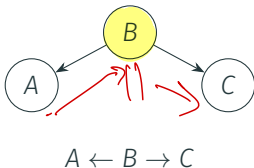
I

Causal chain



$(A \perp C \mid B)$

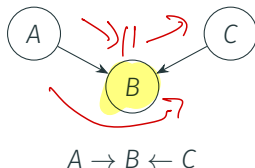
Common cause



$(A \perp C \mid B)$

Specials

Common effect



collider / v-structure

$(A \perp C)$

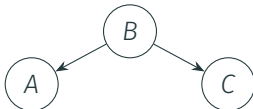
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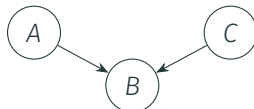
$$A \rightarrow B \rightarrow C$$

Common cause



$$A \leftarrow B \rightarrow C$$

Common effect



$$A \rightarrow B \leftarrow C$$

collider / v-structure

What happens when we observe the middle node B ?

Base Cases: Chain and Common Cause

Causal chain



$$A \rightarrow B \rightarrow C$$

Base Cases: Chain and Common Cause

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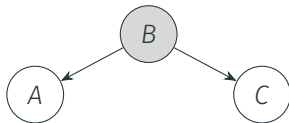


$$A \rightarrow B \rightarrow C$$

Observing B **blocks** the path.

$$A \perp C \mid B$$

Common cause



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Base Cases: Chain and Common Cause

Causal chain

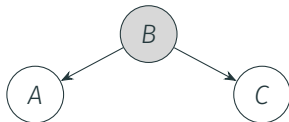


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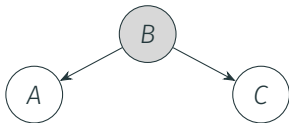


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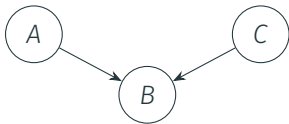
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$$A \perp C \mid B$$

What about a common effect?

Base Case: Common Effect

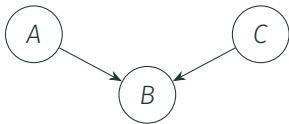
B unobserved



$A \rightarrow B \leftarrow C$

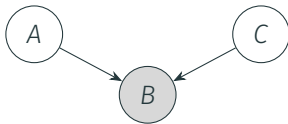
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B observed



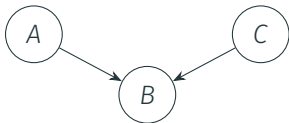
$A \rightarrow B \leftarrow C$

The collider **blocks** the path.

$A \perp C$

Base Case: Common Effect

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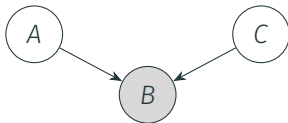


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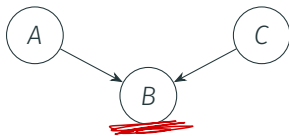
$$A \rightarrow \textcolor{brown}{B} \leftarrow C$$

Observing the collider **opens** the path.

$$A \not\perp C \mid B$$

Base Case: Common Effect

B unobserved

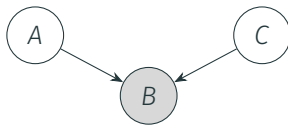


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Colliders behave differently!

d-separation = direction-dependent separation

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- **Motivation**

How is conditional independence in a BN encoded by the structure of its DAG?

d-separation = direction-dependent separation

- Motivation

How is conditional independence in a BN encoded by the structure of its DAG?

- Theorem

$P(X, Y|E) = P(X|E)P(Y|E)$ if and only if every path from a node in X to a node in Y is blocked by E .

d-separation = direction-dependent separation

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How is conditional independence in a BN encoded by the structure of its DAG?

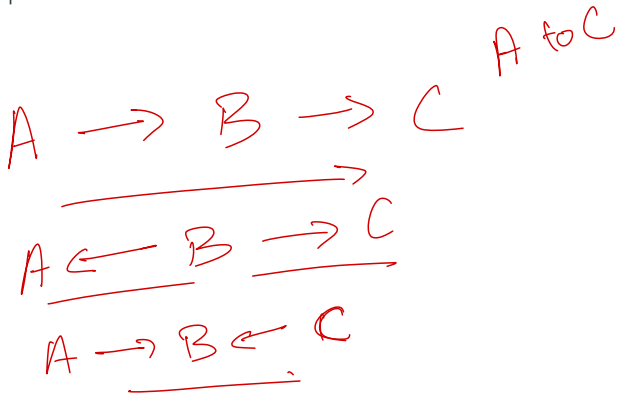
- Theorem

$P(X, Y|E) = P(X|E)P(Y|E)$ if and only if every *path* from a node in X to a node in Y is *blocked* by E .

What counts as a path, and when is it blocked?

- Definition

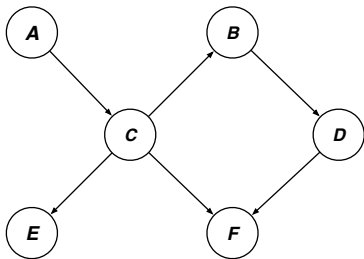
A **path** is any sequence of nodes connected by edges (regardless of their directionalities); it is also assumed that no nodes repeat.



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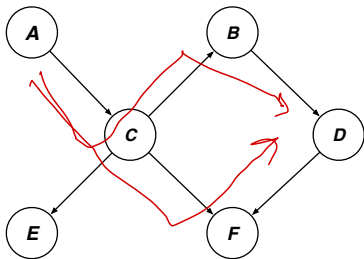


Paths in DAGs

- Definition

A **path** is any sequence of nodes connected by edges (*regardless of their directionalities*); it is also assumed that no nodes repeat.

- Examples



Two ? paths from A to D:

(1) $\underline{A} \rightarrow \underline{C} \rightarrow \underline{B} \rightarrow D$

(2) $\underline{A} \rightarrow \underline{C} \rightarrow \underline{F} \leftarrow D$

Blocked paths

- Definition

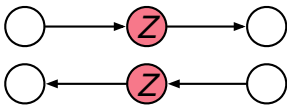
A path π is **blocked** by a set of nodes E if there exists a node $Z \in \pi$ for which one of the three following conditions hold.

Blocked paths

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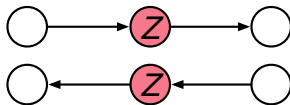
edges **align**

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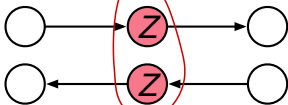
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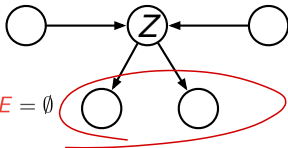
*given in the evidence.
then the path
edges align
is blocked*

(2) $Z \in E$



edges **diverge**

(3) $Z \notin E$



edges **converge**

$\text{descendants}(Z) \cap E = \emptyset$

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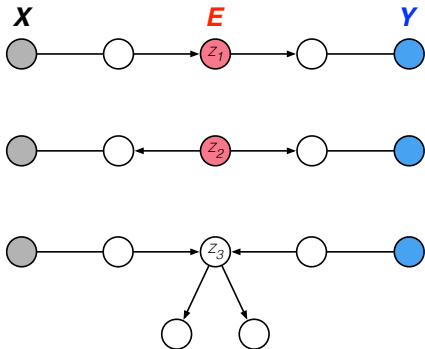
- Intuition

d-separation

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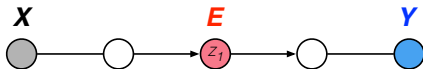


d-separation

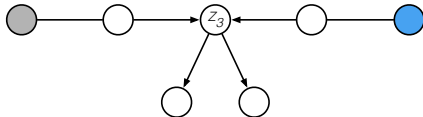
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- Intuition



$Z_1 \in E$ is an intervening event in a causal chain

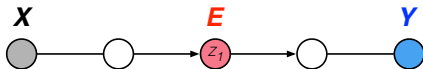


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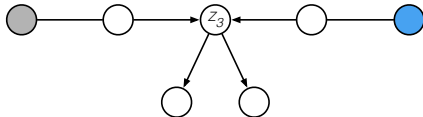
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$Z_1 \in E$ is an intervening event in a causal chain



$Z_2 \in E$ is a common explanation or cause

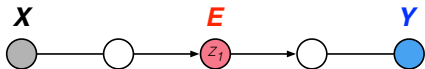


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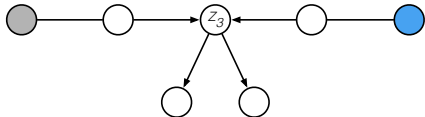
- Intuition



$Z_1 \in E$ is an intervening event in a causal chain



$Z_2 \in E$ is a common explanation or cause



$Z_3 \notin E$, $\text{desc}(Z_3) \cap E = \emptyset$ is an unobserved common effect

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- **Proof** (not given)

- **Theorem**

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- **Proof** (not given)

The proof of the theorem is non-trivial.

You are **not** responsible for its proof.

- Theorem

$P(X, Y|E) = P(X|E)P(Y|E)$ if and only if every *path* from a node in X to a node in Y is *blocked* by E .

- **Proof** (not given)

The proof of the theorem is non-trivial.

You are **not** responsible for its proof.

- How useful is the theorem?

- Theorem

$P(X, Y|E) = P(X|E)P(Y|E)$ if and only if every *path* from a node in X to a node in Y is *blocked* by E .

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The proof of the theorem is non-trivial.

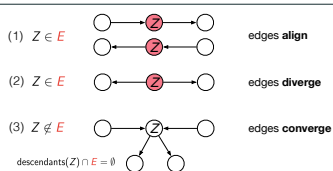
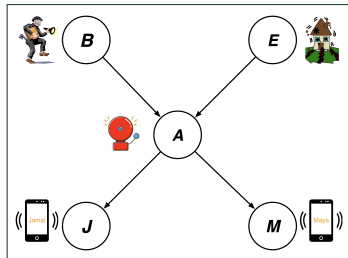
You are **not** responsible for its proof.

- **How useful is the theorem?** **Very!**

There are efficient algorithms to test d-separation in large BNs.
You should become skilled at these tests in simple BNs.

Alarm example

A. TRUE or B. FALSE?

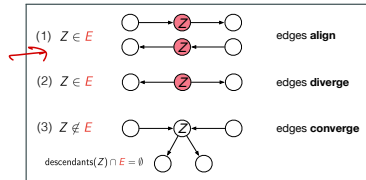
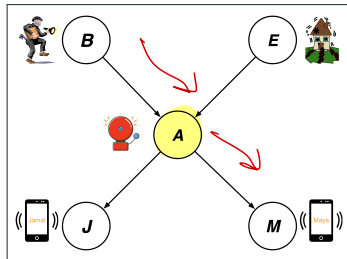


Alarm example

A. TRUE or B. FALSE?

$$1. P(B|A, M) \stackrel{?}{=} P(B|A)$$

if $B \perp M \mid A$
↓ ↓ ↓
X Y E
— —

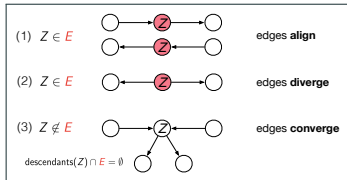
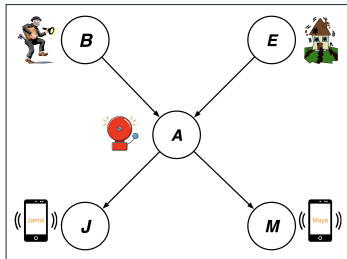


Alarm example

A. TRUE or B. FALSE?

1. $P(B|A, M) \stackrel{?}{=} P(B|A)$

The evidence is $\{A\}$.



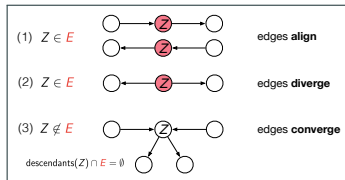
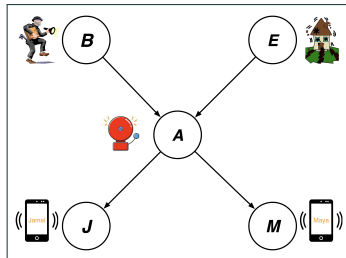
Alarm example

A. TRUE or B. FALSE?

1. $P(B|A, M) \stackrel{?}{=} P(B|A)$

The evidence is $\{A\}$.

There is one path $B \rightarrow A \rightarrow M$.



Alarm example

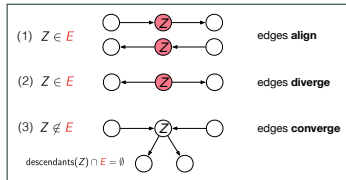
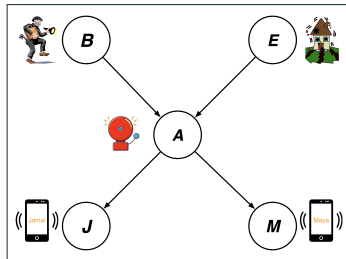
A. TRUE or B. FALSE?

1. $P(B|A, M) \stackrel{?}{=} P(B|A)$

The evidence is $\{A\}$.

There is one path $B \rightarrow A \rightarrow M$.

Node A satisfies condition (1).



Alarm example

A. TRUE or B. FALSE?

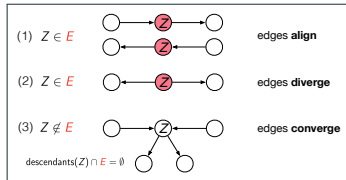
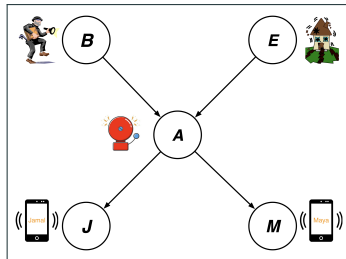
1. $P(B|A, M) \stackrel{?}{=} P(B|A)$

The evidence is $\{A\}$.

There is one path $B \rightarrow A \rightarrow M$.

Node A satisfies condition (1).

The statement is **true**.



Alarm example

A. TRUE or B. FALSE?

$$1. P(B|A, M) \stackrel{?}{=} P(B|A)$$

The evidence is $\{A\}$.

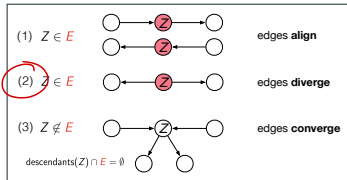
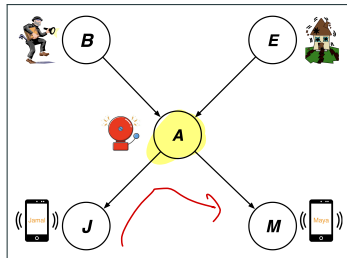
There is one path $B \rightarrow A \rightarrow M$.

Node A satisfies condition (1).

The statement is **true**.

$$2. P(J, M|A) \stackrel{?}{=} P(J|A) P(M|A)$$

$J \perp M | A$?



Alarm example

A. TRUE or B. FALSE?

1. $P(B|A, M) \stackrel{?}{=} P(B|A)$

The evidence is $\{A\}$.

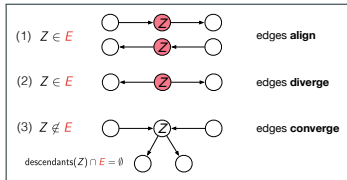
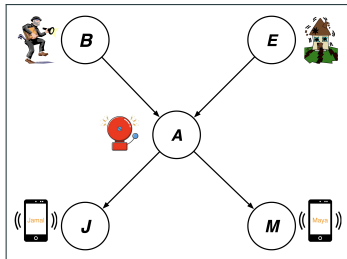
There is one path $B \rightarrow A \rightarrow M$.

Node A satisfies condition (1).

The statement is **true**.

2. $P(J, M|A) \stackrel{?}{=} P(J|A) P(M|A)$

The evidence is $\{A\}$.



Alarm example

A. TRUE or B. FALSE?

1. $P(B|A, M) \stackrel{?}{=} P(B|A)$

The evidence is $\{A\}$.

There is one path $B \rightarrow A \rightarrow M$.

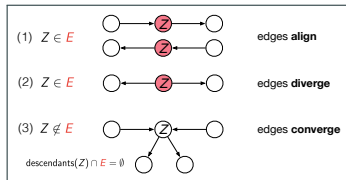
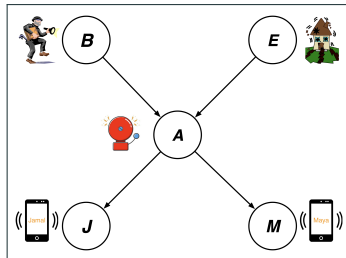
Node A satisfies condition (1).

The statement is **true**.

2. $P(J, M|A) \stackrel{?}{=} P(J|A) P(M|A)$

The evidence is $\{A\}$.

There is one path $J \leftarrow A \rightarrow M$.



Alarm example

A. TRUE or B. FALSE?

1. $P(B|A, M) \stackrel{?}{=} P(B|A)$

The evidence is $\{A\}$.

There is one path $B \rightarrow A \rightarrow M$.

Node A satisfies condition (1).

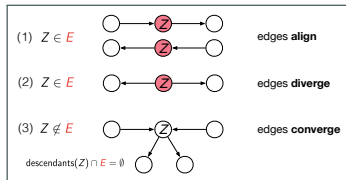
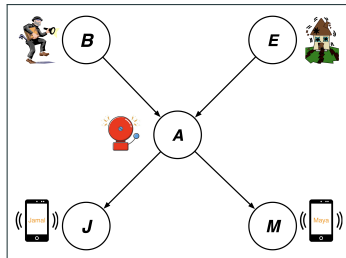
The statement is **true**.

2. $P(J, M|A) \stackrel{?}{=} P(J|A) P(M|A)$

The evidence is $\{A\}$.

There is one path $J \leftarrow A \rightarrow M$.

Node A satisfies condition (2).



Alarm example

A. TRUE or B. FALSE?

1. $P(B|A, M) \stackrel{?}{=} P(B|A)$

The evidence is $\{A\}$.

There is one path $B \rightarrow A \rightarrow M$.

Node A satisfies condition (1).

The statement is **true**.

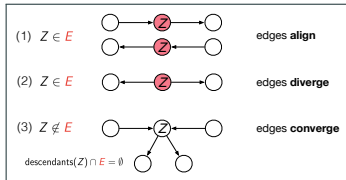
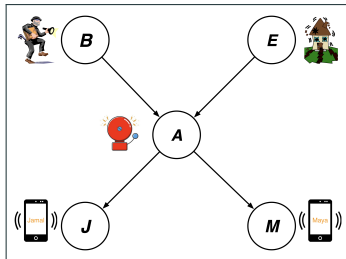
2. $P(J, M|A) \stackrel{?}{=} P(J|A) P(M|A)$

The evidence is $\{A\}$.

There is one path $J \leftarrow A \rightarrow M$.

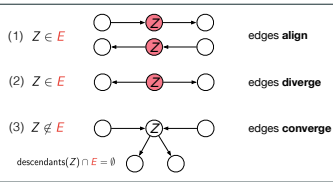
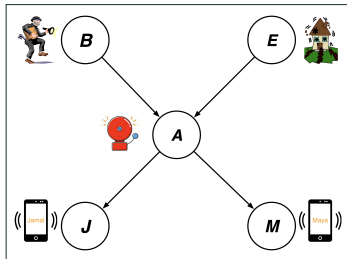
Node A satisfies condition (2).

The statement is **true**.



Alarm example (con't)

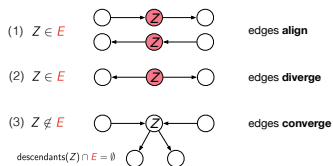
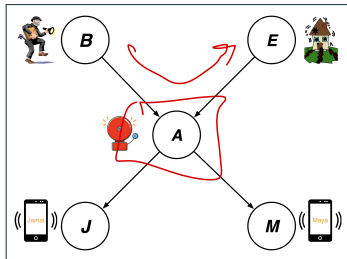
A. TRUE or B. FALSE?



Alarm example (con't)

A. TRUE or B. FALSE?

3. $P(B) \stackrel{?}{=} P(B|E)$

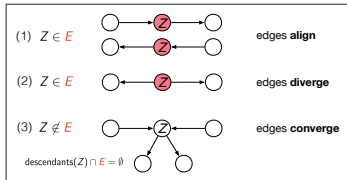
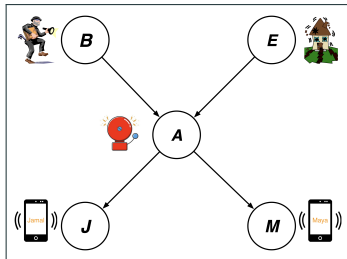


Alarm example (con't)

A. TRUE or B. FALSE?

3. $P(B) \stackrel{?}{=} P(B|E)$

The evidence is {}.



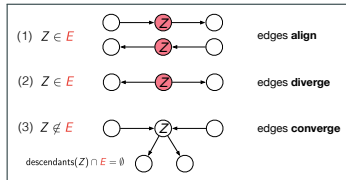
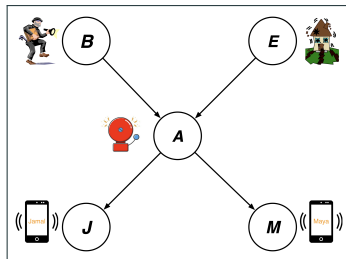
Alarm example (con't)

A. TRUE or B. FALSE?

3. $P(B) \stackrel{?}{=} P(B|E)$

The evidence is $\{ \}$.

There is one path $B \rightarrow A \leftarrow E$.



Alarm example (con't)

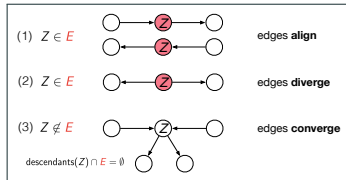
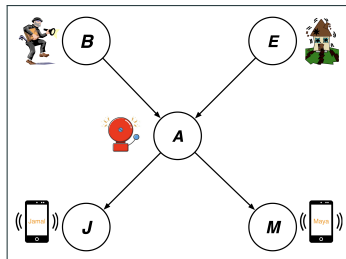
A. TRUE or B. FALSE?

3. $P(B) \stackrel{?}{=} P(B|E)$

The evidence is $\{ \}$.

There is one path $B \rightarrow A \leftarrow E$.

Node A satisfies condition (3).



Alarm example (con't)

A. TRUE or B. FALSE?

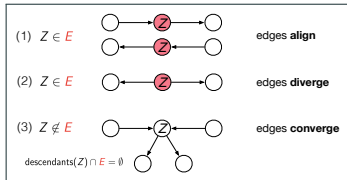
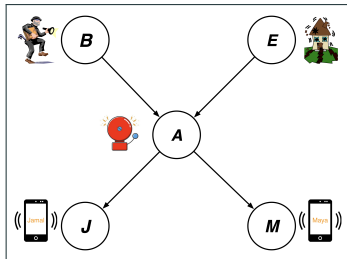
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The evidence is $\{ \}$.

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The statement is **true**.



Alarm example (con't)

A. TRUE or B. FALSE?

3. $P(B) \stackrel{?}{=} P(B|E)$

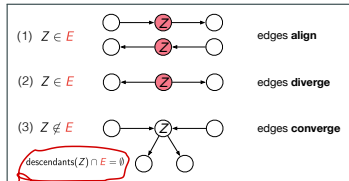
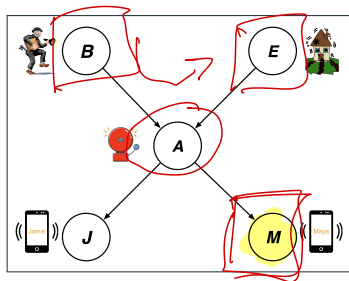
The evidence is $\{ \}$.

There is one path $B \rightarrow A \leftarrow E$.

Node A satisfies condition (3).

The statement is **true**.

4. $P(B|M) \stackrel{?}{=} P(B|M, E)$



~~$J \perp M | E$~~ ?

Alarm example (con't)

A. TRUE or B. FALSE?

3. $P(B) \stackrel{?}{=} P(B|E)$

The evidence is $\{ \}$.

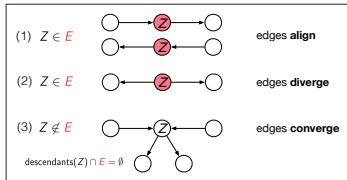
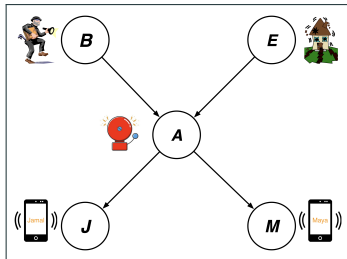
There is one path $B \rightarrow A \leftarrow E$.

Node A satisfies condition (3).

The statement is **true**.

4. $P(B|M) \stackrel{?}{=} P(B|M, E)$

The evidence is $\{M\}$.



Alarm example (con't)

A. TRUE or B. FALSE?

3. $P(B) \stackrel{?}{=} P(B|E)$

The evidence is $\{ \}$.

There is one path $B \rightarrow A \leftarrow E$.

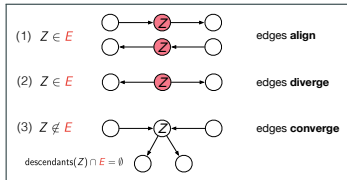
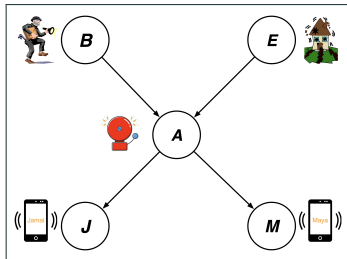
Node A satisfies condition (3).

The statement is **true**.

4. $P(B|M) \stackrel{?}{=} P(B|M, E)$

The evidence is $\{M\}$.

There is one path $B \rightarrow A \leftarrow E$.



Alarm example (con't)

A. TRUE or B. FALSE?

3. $P(B) \stackrel{?}{=} P(B|E)$

The evidence is $\{ \}$.

There is one path $B \rightarrow A \leftarrow E$.

Node A satisfies condition (3).

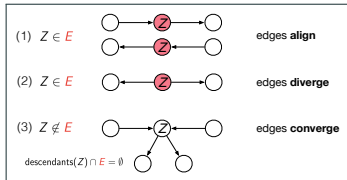
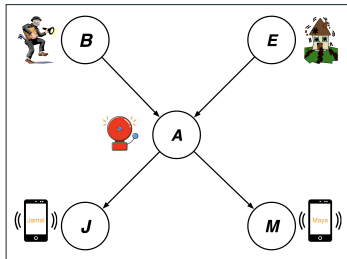
The statement is **true**.

4. $P(B|M) \stackrel{?}{=} P(B|M, E)$

The evidence is $\{M\}$.

There is one path $B \rightarrow A \leftarrow E$.

Note that $M \in \text{desc}(A)$.



Alarm example (con't)

A. TRUE or B. FALSE?

3. $P(B) \stackrel{?}{=} P(B|E)$

The evidence is $\{ \}$.

There is one path $B \rightarrow A \leftarrow E$.

Node A satisfies condition (3).

The statement is **true**.

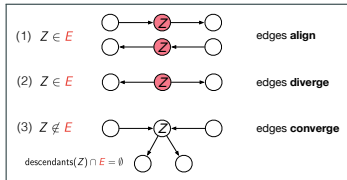
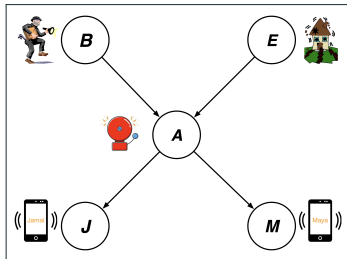
4. $P(B|M) \stackrel{?}{=} P(B|M, E)$

The evidence is $\{M\}$.

There is one path $B \rightarrow A \leftarrow E$.

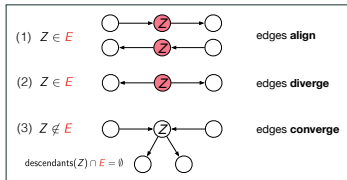
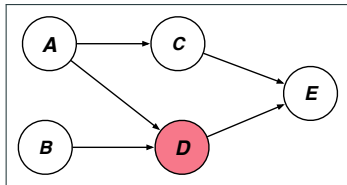
Note that $M \in \text{desc}(A)$.

The statement is **false**.



Loopy example

A. TRUE or B. FALSE?



Loopy example

A. TRUE or B. FALSE?

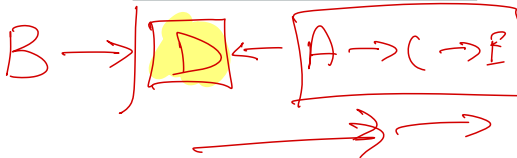
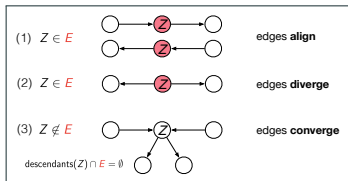
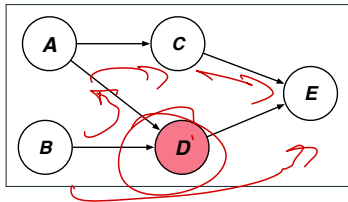
5. $P(B|D, E) \stackrel{?}{=} P(B|D)$

$B \perp E \mid D$

A TRUE $\rightarrow 64\%$

B FALSE $\rightarrow 35\%$

C 1?

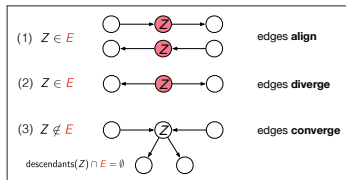
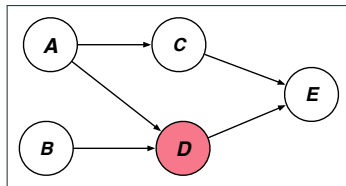


Loopy example

A. TRUE or B. FALSE?

5. $P(B|D, E) \stackrel{?}{=} P(B|D)$

The evidence is $\{D\}$.



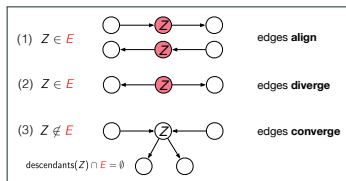
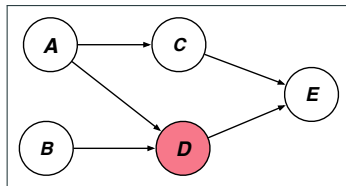
Loopy example

A. TRUE or B. FALSE?

5. $P(B|D, E) \stackrel{?}{=} P(B|D)$

The evidence is $\{D\}$.

There are two paths from B to E .



Loopy example

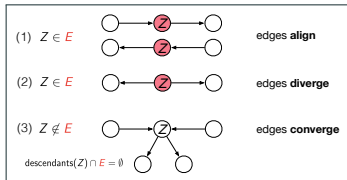
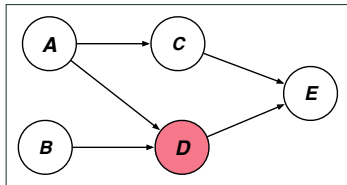
A. TRUE or B. FALSE?

5. $P(B|D, E) \stackrel{?}{=} P(B|D)$

The evidence is $\{D\}$.

There are two paths from B to E .

Path $B \rightarrow D \rightarrow E$
is blocked by node D ,
satisfying condition (1).



Loopy example

A. TRUE or B. FALSE?

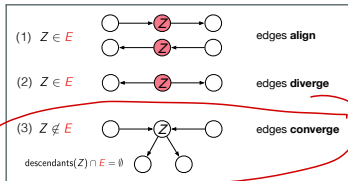
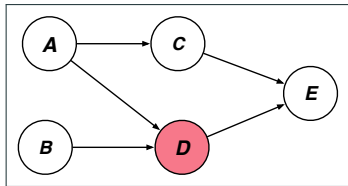
5. $P(B|D, E) \stackrel{?}{=} P(B|D)$

The evidence is $\{D\}$.

There are two paths from B to E .

Path $B \rightarrow \underline{D} \rightarrow E$
is blocked by node D ,
satisfying condition (1).

Path $B \rightarrow \underline{D} \leftarrow A \rightarrow C \rightarrow E$
is not blocked by any node.



Loopy example

A. TRUE or B. FALSE?

5. $P(B|D, E) \stackrel{?}{=} P(B|D)$

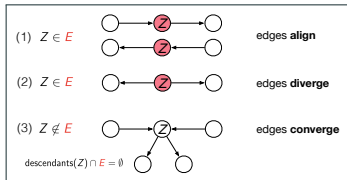
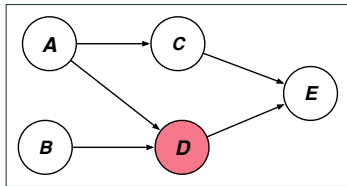
The evidence is $\{D\}$.

There are two paths from B to E .

Path $B \rightarrow D \rightarrow E$
is blocked by node D ,
satisfying condition (1).

Path $B \rightarrow D \leftarrow A \rightarrow C \rightarrow E$
is not blocked by any node.

The statement is **false**.

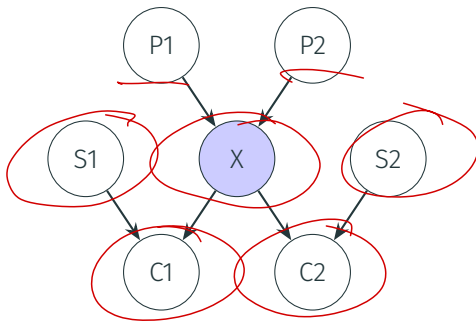


Markov Blanket

A **Markov Blanket** B_X of node X consists of **parents** of X , **children** of X and **"spouses"** (other parents of children of X , but not X) of X .

Markov Blanket

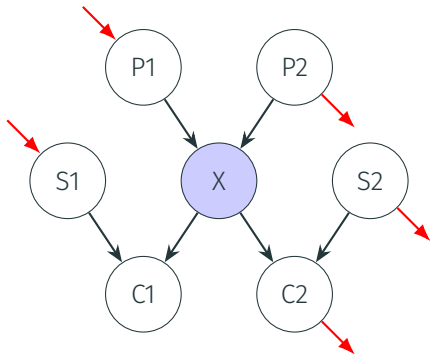
A **Markov Blanket** B_X of node X consists of **parents** of X , **children** of X and **"spouses"** (other parents of children of X , but not X) of X .



Every variable is conditionally independent of any other variable given it's **Markov Blanket**.

Markov Blanket

A **Markov Blanket** B_X of node X consists of **parents** of X , **children** of X and **"spouses"** (other parents of children of X , but not X) of X .



Every variable is conditionally independent of any other variable given it's **Markov Blanket**.

That's all folks!